

# **A Hidden Markov Model to Identify Regions of Interest from Eye Movements, with an Application to Nodule Detection in Chest X-Rays**

MICHEL WEDEL<sup>a1</sup>, JIN YAN<sup>b</sup>, PAUL J. SMITH<sup>b</sup>, ELIOT L. SIEGEL<sup>c</sup>, and HONGSHUAN LI<sup>a</sup>

<sup>a</sup>Robert H. Smith School of Business, University of Maryland, College Park, MD, 20782

<sup>b</sup>Department of Mathematics, University of Maryland, College Park, MD, 20782

<sup>c</sup>University of Maryland School of Medicine, Baltimore, MD, 21201

## **ABSTRACT**

Nodules that may represent lung cancer are often missed in chest X-rays by radiologists. Recording of eye movements during search for nodules provides insights into the search process. We develop a Hierarchical Bayes Hidden Markov Model (HMM) and analyze eye tracking data of sixteen laymen looking at fourteen chest X-rays, of which seven contained a potentially cancerous nodule. We use the luminance of pixels in the X-ray image as prior information on the location of a nodule. Using a reversible jump Markov Chain Monte Carlo algorithm to estimate the HMM enables us to identify the number of regions of interest (ROIs) on each image, as well as their centers, sizes and orientations. In the application in most cases one of the ROIs covers the nodule precisely. Our study thus demonstrates that a HMM analysis of eye movements recorded on laymen may accurately reveal the location of nodules, which has significant implications. The HMM model may be useful in other applications to identify ROIs from eye movement data.

**KEY WORDS:** Eye tracking; Reversible Jump Markov Chain Monte Carlo; Hidden Markov Model; Regions of Interest; Lung Cancer; Radiology.

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<sup>1</sup> \* *Correspondence to:* Michel Wedel, Robert H. Smith School of Business, University of Maryland, College Park, MD 20782 (E- mail: [mwedel@rhsmith.umd.edu](mailto:mwedel@rhsmith.umd.edu)).

## INTRODUCTION

Lung cancer is the primary cause of cancer-related deaths for both men and women and every major ethnic group in the US. Over the next twenty years, one expects its incidence to continue to grow substantially (Lung Cancer Mortality Reduction Act, 2011). In the United States, the five year survival rate of lung cancer is only fifteen percent, but early detection through screening programs could increase that to fifty percent, saving tens of thousands of lives a year (Lung Cancer Mortality Reduction Act, 2011). Chest radiography is the most frequently used screening procedure. However, research suggests that in screening programs as much as three quarters of pulmonary nodules, which can represent or develop into lung cancer, may be overlooked (Austin, Romney & Goldmith, 1992; Nodine & Kundel, 1987). In chest X-ray images these nodules show up as bright roundish areas, but are often exceedingly difficult to detect, because they vary in location, shape, size, and brightness (Krupinski, Roehrig, Dallas & Roehrig, 2003). Radiologists need to process vast amounts of X-ray images within limited amounts of time, a situation that will become even worse with the implementation of large scale screening programs. Therefore, tools to assist radiologists in searching for nodules in X-ray images and to increase the efficiency and effectiveness of the detection of nodules will become ever more important. Medical image processing technologies hold promise to support improved detection, but a problem is currently still their high false positive rates, which, because false positives need to be inspected by radiologists, may slow down, rather than speed up the search process (Kakeda et al., 2004).

A body of research has therefore sought to better understand the search skills of radiologists, with the aim of improving computerized processing and display of medical

images, and the efficiency and effectiveness of radiologists' search for nodules through training and feedback. Eye movements made during visual search may provide insights into the search processes that cannot be otherwise easily obtained (Krupinski, 2000). Recent advances in technology make it possible to record eye movements unobtrusively, accurately, and at relatively low cost (Duchowski, 2003). Several studies have used eye movement data to investigate whether visual search patterns can reveal the nature of errors (Nodine & Kundel, 1987), which image and nodule characteristics contribute to correct and incorrect detections (Oestman et al., 1988; Krupinski et al., 2003), and what the causes are for the high variability in the ability of radiologists to detect nodules (Krupinski, 2000).

In this study we propose a Bayesian Hidden Markov Model (HMM; Rabiner, 1990; Scott, 2002) to investigate the search process for nodules in chest X-rays and help identify the location of a nodule from eye movements. A HMM is a probabilistic model to describe patterns in longitudinal data: the observed data is assumed to reflect a set of unobserved states, and knowledge of the behavior of these states is of key interest. Each of the hidden states is assumed to produce a unique pattern in the observed data. The transitions between the hidden states are assumed to follow a Markov process, that is, the probability of the hidden state at time  $t$  is only dependent on the hidden state at the previous time point,  $t-1$ . The HMM is widely used in a variety of fields of research (Scott, 2002), and has been previously used for the description of eye movement data as well (Liechty, Pieters, & Wedel 2003; Wedel, Pieters, & Liechty 2008; Van der Lans, Pieters & Wedel, 2008). Liechty, Pieters and Wedel (2003) developed a two-state continuous time HMM, that, through constraints on length of saccades in each state, allowed them to identify local and

global patterns of visual exploration on static images. Wedel, Liechty and Pieters (2008) apply a similar HMM to show how the people's goals affect their local and global exploration of visual stimuli. Van der Lans, Pieters and Wedel (2008) developed a two-state HMM in which the eye movements in the two states were parameterized differently, which allowed them to distinguish a localization state and an identification state during visual search.

In this study, we use the HMM to identify Regions of Interest (ROI) on X-ray images. Much of eye tracking research relies on such Regions of Interest, used to classify eye movements (Holmqvist et al., 2011). The definition of ROIs allows eye tracking researchers to investigate what aspects of an image respondents inspect, by analyzing how long they dwell within each ROI, and how they move from one ROI to another. The locations and shapes of ROIs are mostly defined a- priori based on hypotheses of the researcher, and software packages allow eye tracking researchers to manually define ROIs on their stimuli (Holmqvist et al., 2011). In contrast, we will use the HMM model to identify the ROIs from the eye tracking data itself. The hidden states in the HMM represent the ROIs. The observed eye fixations of each participant on an image are assumed to arise from a two-dimensional spatial Poisson process and to cluster in the ROIs, each of which might potentially contain a nodule. Except for one, the headline or image caption, these ROIs are unknown in number, position, size and orientation. In the HMM model, the scan-path of the eyes across the unobserved ROIs is assumed to follow a first order Markov process. The states of the Markov process are thus partially hidden, as only the first state (image caption) is observed.

We formulate and estimate the HMM in the Bayesian statistical framework, which provides a formalized way of formulating a model and learning about its parameters (Lee, 2011). A prior distribution is formulated for these parameters, which is updated with information contained in the data as captured by the likelihood. This leads to the posterior distribution of the parameters, which may be complex but can often be conveniently factored in such a way that Markov Chain Monte Carlo (MCMC) methods can be used for inference. MCMC simulation from the posterior distribution involves recursively sampling from a series of known full conditional posterior distributions, and summarizing the samples of parameter values thus obtained (Wedel & Lenk, 2013). The Bayesian framework allows us to accommodate unobserved differences between participants in the transitions between the hidden states. In addition, it allows us to use the luminance of pixels in the image as a prior on the location of the ROIs, which should help their identification because nodules are usually brighter than the background. Further, unlike previous applications of the HMM to eye movement data, we do not assume the number of hidden states (ROIs) known, but we estimate it from the eye movement data, using a Reversible Jump MCMC algorithm (Green, 1995).

We analyze data from an eye tracking experiment in which sixteen participants search for nodules on X-ray images. The participants in the study are laypeople that are not trained in the detection of nodules. Previous research has shown that laymen and trained radiologists do not fundamentally differ in their visual skills (Nodine & Krupinski, 1998), and using laymen provides a conservative test for the performance of the HMM. We are interested in characterizing and understanding the search process on the unobserved ROIs, with the aim of providing information that can help in the training of radiologists and the

better detection of nodules. Section 2 describes the data collected on participants' eye movements while searching for nodules in fourteen images of which seven contained a nodule. Section 3 describes the Hidden Markov Model to identify ROIs from eye movement data, and Section 4 describes the results, showing that the combination of the proposed HMM and eye tracking data can reliably detect ROIs that contain nodules in X-ray images. We will reflect on the implications in Section 5.

## DATA

The data were collected on participants in the eye tracking lab of the University Maryland. The participants are undergraduate students in business who volunteered to participate for course credit. They were instructed about the purpose and duration of the study. The instructions and images were presented on 21-inch LCD monitors and were shown full screen at a resolution of  $1,024 \times 768$  pixels (approximately  $15.5^\circ \times 19.4^\circ$  of visual angle). Participants seated in front of the monitor were first exposed to an annotated X-ray image of a chest to familiarize themselves with various structures (heart, trachea, hilum, diaphragm, spine, ribs and lungs). The X-ray images were shown in their original grayscales. Then, to provide them with some training in finding nodules, participants were exposed to fourteen X-ray images of which seven did and seven did not contain a potentially cancerous nodule. They were instructed to search for a possible nodule, present as a bright round shape in the image, and take as much time as needed for each image. After their search of an image was complete, they were provided with feedback on the presence and location of the nodule.

Next, participants were exposed to fourteen testing X-ray images and were asked to search for a possible nodule on the lungs. Among the fourteen X-ray images, seven contained a single nodule and seven did not. Participants did not know this information. Figure 1 shows an example of an X-ray image with a nodule present. Each image contained a headline or caption, shown on top, which contained brief demographic and symptomatic information. Notice from Figure 1 that the nodule is very difficult to detect, which is even the case for trained radiologists. The images were presented in random order and participants were asked to search for a single nodule on each of them. During their search of each image, participants' eye movements were recorded with infrared corneal reflection eye tracking methodology. The eye tracker (Tobii® 1750) registers the point of regard every 50 milliseconds with a spatial resolution better than 0.5° and measures the reflection of infrared rays emitted by the eye-tracker on the cornea (the hard outer layer) of both eyes. The miniature infrared cameras are embedded in the rim of the monitor on which the images are displayed. The eye tracking equipment allows participants to freely move their heads within a virtual box of about 20 inches, while miniature cameras track the position of the head and eyes.

The two main types of eye movements are fixations and saccades (Rayner, 1998). During an eye fixation the eye is relatively still for 100-500 milliseconds. Visual information around the point of regard is extracted and projected onto the fovea. This is a small area of the retina with high acuity, covering about two degrees of visual angle. A saccade is a ballistic movement in the range of 20-50 milliseconds during which the eye is rapidly redirected to potentially informative regions on the image. During a saccade vision is suppressed. A sequence of recorded fixations and saccades on an image is called a scan-

path. In the present study, fixation locations of individual participants, measured in pixel coordinates on the image, are the unit of analysis. From the raw point-of-regard data produced by the eye tracking equipment, eye fixation positions were computed with the algorithm of Van der Lans, Wedel and Pieters (2011). Participants who had less than forty points-of-regard recorded during exposure were removed from the study. This resulted in a final set of sixteen participants with usable data (one participant missing Case 11). When the participants believed they had reached a decision on the presence of a nodule in an image, they were asked to fixate the location in question for about two seconds, and click on a button on the bottom right of the screen to proceed to the next image. They were also asked to record whether they believed whether there is a nodule. In the present study, the fixation sequences of sixteen participants on each of seven separate X-ray images which have a nodule present will be used for analysis (Cases 1, 3, 9, 10, 11, 12, and 14).

[Insert Figure 1 about here]

## HIDDEN MARKOV MODEL

In this section we describe the HMM used to identify ROIs on a given X-ray image from participants' eye tracking data. Note in Figure 1 that the shape and location of various structures present in the image is complex and varies between images, which makes the a-priori definition of ROIs difficult. In formulating the HMM, the ROIs are assumed to be common to all participants, but their number is assumed unknown, and they are assumed to have unknown centers, sizes and orientations, except for the first one, which is the image caption. Participants' eyes move among these unobserved ROIs when searching for a

nodule on the X-ray image in question, according to a Markov process; each participant having a different transition matrix. The next sections, respectively, define the Hidden Markov Model, provide the likelihood and the posterior distributions, and describe the details of the Reversible Jump Markov Chain Monte Carlo (MCMC) algorithm.

We assume that, regardless of participants and time, a given image contains  $S$  ROIs, which may partly overlap. These ROIs are the states in a Hidden Markov Chain. The data of participant  $i$  consists of a sequence of pairs  $(\vec{y}_{i,t}, s_{i,t})$  with length  $T_i$ , where  $\vec{y}_{i,t}$  is a two-dimensional vector representing the fixation point at time  $t$ , and  $s_{i,t}$  is the corresponding ROI. We let  $\mathbf{y}_i = \{\vec{y}_{i,1}, \dots, \vec{y}_{i,T_i}\}$  denote the fixation sequence for participant  $i$  and  $\mathbf{s}_i = \{s_{i,1}, \dots, s_{i,T_i}\}$  the sequence of ROIs visited. For participant  $i$  at time  $t$ ,  $(\vec{y}_{i,t}, s_{i,t})$  is uniquely defined, but only  $\mathbf{y}_i$  is observable.

### Spatial Poisson Process

Conditional on  $s_{i,t}$ , we assume that the eye fixation positions  $\vec{y}_{i,t}$  are independently distributed according to a spatial Poisson process, with intensity function

$$\lambda_s(\vec{y}) = \lambda_s(\vec{y}_{i,t} = \vec{y} | s_{i,t} = s) = \mathcal{N}_2(\vec{y} | \vec{\mu}_s, \Sigma_s) I(\vec{y} \in \Omega), \quad s = 1, \dots, S, \quad (1)$$

where  $\mathcal{N}_2$  denotes a bivariate normal density,  $\Omega$  denotes collection of pixels on the image,  $\vec{\mu}_s = (\mu_s^x, \mu_s^y)'$  is the spatial center of the  $s^{\text{th}}$  ROI and  $\Sigma_s$  its spatial covariance matrix. Thus, the unobserved ROIs are assumed to have the shape of ellipses constrained to the region of the image. Different values of the parameters of the bivariate Normal distributions allow for a broad range of locations and shapes of the ROIs. By the property of spatial Poisson process, the probability of the eye fixating at point  $\vec{y}_{i,t}$  conditional on  $s_{i,t}$  is:

$$P(\vec{y}) = P(\vec{y}_{i,t} = \vec{y} | s_{i,t} = s) = \frac{\lambda_s(\vec{y})}{\int_{\Omega} \lambda_s(\vec{u}) d\vec{u}} = \frac{\mathcal{N}_2(\vec{y} | \vec{\mu}_s, \Sigma_s) I(\vec{y} \in \Omega)}{\int_{\Omega} \mathcal{N}_2(\vec{u} | \vec{\mu}_s, \Sigma_s) d\vec{u}}, \quad (2)$$

where  $\vec{u} = (u_x, u_y) \in \Omega$ , and both  $\vec{y}_{i,t}$  and  $\vec{u}$  are measured in image pixel units in the application.

### Spatial and Luminance Priors

On X-ray images, independent of ambient and monitor lighting, nodules are visible as roundish regions that tend to be brighter than the background. A region of relatively brighter pixels is thus a-priori more likely to contain a nodule. Therefore, apart from spatial information, pixel luminance should a priori influence the probability that a nodule is present, where higher luminance corresponds to a larger probability of a nodule. This is also the basic idea underlying medical image processing techniques. For the luminance prior, we choose the following form:

$$P_{lu}(\vec{\mu}_s) \propto (\beta + 1)(x_{\vec{\mu}_s})^{\beta}, \quad (3)$$

where  $x_{\vec{\mu}_s} \in [0,1]$  is the luminance value at  $\vec{\mu}_s$ . This luminance value is computed with standard image processing methods as the grayscale intensity of that pixel, measured in  $[0,1]$ . The higher the value, the brighter the pixel is. The parameter  $\beta$  has a non-informative uniform hyper prior  $P(\beta) = U(-1,1)$ . If the posterior of  $\beta$  has a substantial part of its mass above 0, it means that the brighter, more luminous points are more likely to be the center of the ROIs. In this way, the proposed approach combines information in the image with the information contained in respondents' eye movements to identify ROIs.

We assign both spatial and luminance priors to the locations of the ROIs. For  $s = 1, \dots, S$ , the spatial prior for  $\{\vec{\mu}_s, \Sigma_s\}$  is the Normal-Inverse-Wishart distribution:

$$P_{sp}(\vec{\mu}_s, \Sigma_s) = \mathcal{N}_2(\vec{\mu}_s | m_s, \lambda_s^{-1} \Sigma_s) I(\vec{\mu}_s \in \Omega) \times \mathcal{W}_2^{-1}(\Sigma_s | V_s, \kappa_s), \quad (4)$$

where  $\mathcal{W}_2^{-1}$  denotes the two-dimensional inverse Wishart distribution, and  $m_s$ ,  $\lambda_s$ ,  $V_s$  and  $\kappa_s$  are hyper-parameters; the support of the distribution is restricted to be on the image.

The full prior distribution for the parameters describing the location and shape of the unobserved ROIs is then proportional to:

$$P(\vec{\mu}_s, \Sigma_s) \propto P_{sp}(\vec{\mu}_s, \Sigma_s) \times P_{lu}(\vec{\mu}_s). \quad (5)$$

### Partially Hidden Markov Chain

Instead of using a Markov Chain in which all states are unobserved, we assume that the first state represents the image caption. Such a caption is present at the top of each X-ray image, so that for any fixations  $\vec{y}_{i,t}$  on the caption we have  $s_{i,t} = 1$ . Because the location and shape of the caption is precisely known, the first state in the Markov Chain is observed. There are several reasons for this formulation. First, the headline can never contain the nodule, and therefore can be treated as an a priori known and predefined ROI. Second, it is of interest to investigate how participants jump between the headline and the other ROIs in the image, for example to form an initial or confirm a final judgment. Specifying the headline as an a priori defined ROI makes this possible. Because the first state is observed but the others are not, the model is a Partially Hidden Markov Model.

The sequence  $\mathbf{s}_i = \{s_{i,1}, \dots, s_{i,T_i}\}$  is a partially hidden chain for participant  $i$ , and each  $\mathbf{s}_i$  is assumed to follow a first order time homogeneous Markov chain with transition

matrix  $Q_i = (q_i(r, s))$ , and initial distribution  $\vec{\pi}_{0,i} = \{\pi_{0,i}(1), \dots, \pi_{0,i}(S)\}$ , where for  $s, r \in \{1, 2, \dots, S\}$  and  $t = 2, \dots, T_i$ ,  $q_i(r, s) = p(s_{i,t} = s | s_{i,t-1} = r)$  and  $\pi_{0,i}(s) = p(s_{i,1} = s)$ . For any  $r = 1, \dots, S$ ,  $q(r, \cdot)$ , the  $r^{\text{th}}$  row of  $Q_i$  and the parameters of the initial distribution,  $\vec{\pi}_{0,i}$ , are assumed to be a priori independently Dirichlet distributed:  $D_S(\vec{\alpha}_i)$ .

In order to accommodate unobserved differences among participants in the transition patterns of their eyes among the ROIs, we assume  $\vec{\alpha}_i = A_i \vec{\varphi}_i$ , where  $A_i = \sum_j \alpha_i(j)$  is a shrinkage parameter and  $\vec{\varphi}_i \sim D(1)$  is a location parameter (Scott, James, & Sugar, 2005). The  $A_i$  represent the number of prior observations in the distribution of  $Q_i$  and is given a uniform prior;  $D$  denotes the Dirichlet distribution. The transformed prior for  $\vec{\alpha}_i$  is then:

$$P(\vec{\alpha}_i) = \frac{z_0 \Gamma(S)}{(z_0 + A_i)^2 A_i^{S-1}}, i = 1, \dots, I, \quad (6)$$

with  $\Gamma$  the gamma function. The total number of states  $S$  is unknown, and we assume that a priori it is equally likely to take on any value between 1 and  $S_{\max}$ :  $P(S = k) = \frac{1}{S_{\max}}$ . In order to avoid label switching (Scott, 2002), we order the states of the HMM, except for the first one, by their x-coordinates.

## MCMC ALGORITHM

We estimate the HMM outlined above through the application of Bayesian methods. The Bayesian statistical framework provides a formalized way of formulating a model and learning about its parameters, by updating prior knowledge on these parameters with information contained in the data as captured by the likelihood. It addresses many of the

computational challenges of analyzing eye movement data and presents a very flexible framework for developing and estimating models (Lee, 2011). The Bayesian approach accommodates finite samples, valid distributional assumptions, individual differences, latent variables, and hierarchical data structures, and allows for theoretically more sound testing of hypotheses (Kruschke, 2013). It enables probabilistic models to be developed that represent multiple components of underlying cognitive processes, such as the HMM specified above, and to be estimated on a single set of observed eye movement data using Markov Chain Monte Carlo (MCMC) methods (Wedel & Lenk, 2013). For our particular model, MCMC allows us to estimate the HMM with both unobserved individual differences in the transition probabilities between hidden states, as well as an unknown number of hidden states (ROIs). Estimation of models with MCMC operates through recursive simulation from the posterior distributions of model parameters, which are derived for our model in the next sections.

### Likelihood and Posterior Distributions

The model parameters are  $\Theta = \{\vec{\mu}_{1:S}, \Sigma_{1:S}, Q_{1:I}, \vec{a}_{1:I}, \vec{\pi}_{0,1:I}, \beta, S\}$ , the hyper parameters are

$\Phi = \{\vec{m}_{1:S}, \lambda_{1:S}, V_{1:S}, \kappa_{1:S}, z_0\}$ . The observed data are  $\mathbf{y}_i = \{\vec{y}_{i,1:T_i}\}$ , the latent data are

$\mathbf{s}_i = \{s_{i,T_i}\}$ , and  $X = \{x_{\vec{u}}, \vec{u} \in \Omega\}$  is the luminance at each pixel point on the image. We

define the following:  $n_{0,i}(s) = I(s_{i,1} = s)$ ,  $n_i(s, r) = \sum_{t=2}^{T_i} I(s_{i,t-1} = s, s_{i,t} = r)$ ,

$c_s = \sum_{i=1}^I \sum_{t=1}^{T_i} I(s_{i,t} = s)$ ,  $\vec{a}_s = c_s^{-1} \sum_{i=1}^I \sum_{t:s_{i,t}=s} \vec{y}_{i,t}$ ,  $B_s^2 = \sum_{i=1}^I \sum_{t:s_{i,t}=s} (\vec{y}_{i,t} - \vec{a}_s)(\vec{y}_{i,t} - \vec{a}_s)'$ ,

$M_s = (\lambda_s + c_s)^{-1}(\lambda_s \vec{m}_s + c_s \vec{a}_s)$ ,  $G_s = (\lambda_s + c_s)^{-1} \Sigma_s$ , and  $D_s = \{V_s^{-1} + B_s^2 + \lambda_s c_s(\lambda_s +$

$c_s)^{-1}(\vec{a}_s - \vec{m}_s)(\vec{a}_s - \vec{m}_s)', \kappa_s + c_s\}$ . Then, the joint likelihood of the data for participant  $i$  is:

$$\begin{aligned}
P(\mathbf{y}_i, \mathbf{s}_i, S, \Theta_s, \phi | X) &= \frac{1}{2S} \left\{ \left( \prod_{s=1}^S \pi_{0,i}(s)^{n_{0,i}(s)} \right) D_s(\pi_{0,i} | \vec{\alpha}_i) \right. \\
&\times \left. \prod_{s=1}^S \left[ \left( \prod_{r=1}^S q_i(s, r)^{n_{i(s,r)}} \right) D_s(q_i(s, \cdot) | \vec{\alpha}_i) \right] \right\} \\
&\times \prod_{s=1}^S \left[ \frac{\mathcal{N}_2(\vec{\mu}_s | M_s, G_s) \mathcal{W}_2^{-1}(\Sigma_s | D_s) (\vec{x}_{\vec{\mu}_s})^\beta}{\int_{\vec{u} \in \Omega} \mathcal{N}_2(\vec{u} | \vec{\mu}_s, \Sigma_s) d\vec{u}} \right] \prod_{i=1}^I \left[ \frac{z_0 \Gamma(S)}{(z_0 + A_i)^2 (A_i)^{S-1}} \right] (\beta + 1)^S. \quad (7)
\end{aligned}$$

The following prior settings are used:  $\vec{m}_1 = (460, 60)'$  for the image caption, the image center  $\vec{m}_s = (460, 300)'$  for all other  $s$ ,  $V_s = \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$ ,  $\lambda_s = 0.005$ , and  $\kappa_s = 5$  for all  $s$ . We set  $z_0 = 1$  (some experimentation with this value showed that small perturbations of  $z_0$  had little impact on the posterior distribution; Scott, James & Sugar, 2005).

The parameters  $\Theta$  are sampled from their posterior distributions using an MCMC algorithm. This involves sampling recursively from  $P(\Theta_s | \mathbf{y}_i, \mathbf{s}_i, S, \phi, X)$ , which is factored into four full conditional posterior distributions for which Gibbs sampling or Metropolis-Hasting sampling algorithms are used (Wedel & Lenk, 2013):

$$P(\vec{\alpha}_i | \cdot) \propto \frac{\Gamma(A_i)}{(z_0 + A_i)^2 (A_i)^{S-1}} \prod_{s=0}^S \prod_{r=1}^S \frac{q_i(s, r)^{\alpha_i(r)-1}}{\Gamma(\alpha_i(r))}, \quad (8)$$

$$P(\vec{\mu}_s, \Sigma_s | \cdot) \propto \frac{\mathcal{N}_2(\vec{\mu}_s | M_s, G_s) \mathcal{W}_2^{-1}(\Sigma_s | D_s) (\vec{x}_{\vec{\mu}_s})^\beta}{\left[ \int_{\vec{u} \in \Omega} \mathcal{N}_2(\vec{u} | \vec{\mu}_s, \Sigma_s) d\vec{u} \right]^{c_s}}, \quad (9)$$

$$P(\vec{\pi}_{0,i} | \cdot) \propto D_s(\vec{\pi}_{0,i} | \vec{\alpha}_i + \vec{n}_{0,i}), \quad (10)$$

$$P(q_i(s, \cdot) | \cdot) \propto D_S(q_i(s, \cdot) | \vec{\alpha}_i + \vec{n}_i(s, \cdot)), \quad (11)$$

$$P(\beta | \cdot) \propto (\beta + 1)^S \prod_{s=1}^S (x_{\vec{\mu}_s})^\beta, \quad (12)$$

Further, the MCMC algorithm involves sampling from  $P(\mathbf{s}_i | \mathbf{y}_i, S, \Theta_s \phi, X)$ , utilizing the forward- backward algorithm (Scott, 2002), and from  $P(S | \mathbf{y}_i, \mathbf{s}_i, \Theta_s, \phi, X)$ , for which a reversible jump algorithm is used, as explained below (Green, 1995).

### Reversible Jump Markov Chain Monte Carlo

In our model we assume that the number of ROIs, or states in the HMM, is unknown and thus it must be estimated. This is a difficult inferential problem, for which a special case of Reversible Jump Markov Chain Monte Carlo is used (Green, 1995; Robert, Rydn & Titterington, 2000). The idea is that at each iteration of the MCMC chain, after updating all the hidden states and variables as explained above,  $S$  is randomly increased or decreased by one state, i.e.  $S^* = S + 1$  or  $S^* = S - 1$ , and the current hidden states are rearranged accordingly. Then, a Metropolis-Hastings step is used to determine whether to accept this new number of states,  $S^*$ , or not. The key is that generating or eliminating states should be a reversible pair.

Richardson and Green (1997) describe two reversible jump algorithms. One involves the birth or death of an empty state, while the second one involves a split or merge of existing states. Both of these steps, however, are known to have very low acceptance rates (Robert, Rydn & Titterington, 2000). Our algorithm is a modified birth-death algorithm. That is, in each step of the MCMC chain if the current number of states is  $S$ , the

birth or a death of one state is randomly proposed, with a birth rate  $b_S$  and a death rate  $d_S = 1 - b_S$ . Notice that  $d_1 = 0$ , because the death of a state can only happen when there is at least one empty state. If the current number of states is  $S$  and a birth is proposed, a new state  $j^*$  is generated with a center and a variance covariance matrix  $(\vec{\mu}_{j^*}, \Sigma_{j^*})$  drawn from the prior distribution  $\mathcal{N}_2(\vec{\mu}|m_0, \lambda_0^{-1}\Sigma_0)I(\vec{\mu} \in \Omega)\mathcal{W}_2^{-1}(\Sigma|V_0, \kappa_0)$ . The value of  $j^*$  is determined by the rank of its  $x$ -coordinate and other state indices are reassigned accordingly. The parameter vector  $\vec{\alpha}_i$  is also changed into a  $S + 1$  dimensional vector  $\tilde{\alpha}_i$  by inserting  $\alpha_i(j^*) \sim (S - 1)[A_i + \alpha_i(j^*)]^{1-S}A^{S-1}$ . The transition matrix  $Q_i$  is changed to  $\tilde{Q}_i$  according to:

$$\tilde{q}_i(j, j^*) = u_i(j), \quad (13)$$

$$\tilde{q}_i(j, r) = [1 - u_i(j)]q_i(j, r), \quad (14)$$

for any  $j \neq j^*, r \neq j^*$ , and  $j = 1, \dots, S$ . We use a beta distribution  $B(\alpha_i(j^*), A_i + d_{i,j} - S + 1)$  as the proposal function to generate  $u_i(j)$ . Here  $d_{ij} = \sum_{t=1}^{T_i-1} I(s_{i,t} = j)$ . The new row of  $\tilde{Q}_i$ ,  $\tilde{q}_i(j^*, \cdot)$ , is generated from a  $k + 1$  dimensional Dirichlet distribution  $D_{k+1}(\tilde{\alpha}_i)$ . At this step  $\beta$  does not change. Finally, this birth is accepted with a probability:

$$\min\left(1, R = \frac{\pi(d\tilde{\theta})q(\tilde{\theta}, d\theta)}{\pi(d\theta)q(\theta, d\tilde{\theta})}\right). \quad (15)$$

If the death of a state is proposed, one empty state  $j^*$  is randomly eliminated and the indices are reordered accordingly. The whole  $j^*$ -th row and column of  $Q_i$  are deleted and each row is normalized to sum to one. The acceptance probability is  $\min\{1, R^{-1}\}$ , with  $R$  defined in equation (15). In the empirical application, to a priori not favor a particular

number of states, we use a birth rate  $b_k = 0.5$  for  $s = 2, \dots, S - 1$ ,  $b_1 = 1$  and  $b_S = 0$ . We also use  $S_{\max} = 14$  (the number of states never reached 13 in any of the estimations).

To overcome the problem of a low acceptance rate of births, we add an additional birth step at each iteration of the MCMC chain, with a probability of birth that converges in probability to 0 as the number of iterations goes to infinity. That is, at the  $n^{\text{th}}$  iteration of the MCMC chain, the additional birth occurs with probability  $P_b(n)$ , and if a birth is determined, the exact same birth process as was described above is followed, but that birth is accepted with probability 1. It needs to hold that  $P_b(n) \rightarrow 0$  in probability as  $n \rightarrow \infty$ . Thus, asymptotically this MCMC chain converges to the desired MCMC chain but mixes substantially better during the burn-in period. In the application below, we set  $P_b(n) = 0.005$ ,  $n < 2000$ , and  $P_b(n) = \frac{20,000}{n^2}$ ,  $n \geq 2000$ . The MCMC algorithm was programmed in R. In the application, we use 10,000 draws after a burn-in of 30,000 and thin the draws 1 in 10. Convergence is checked by standard methods, using multiple starting points for the chains.

## Synthetic data analysis

To investigate the performance of our model and algorithm, we analyzed two simulated data sets. We are interested in particular in the recovery of the number of hidden states with our modified reversible jump MCMC algorithm. Each data set was generated assuming four states. In each state the observed data was generated from a spatial Poisson process with a bivariate log-normal intensity. In the first data set, the states were well separated by setting the elements of the variance-covariance matrices to be small, while in second data set the elements of the variance-covariance matrix were large, which resulted

in substantial overlap between the states. The transitions between the four latent states were generated according to a Markov chain. The MCMC chains mixed well during the burn-in, and the algorithm quickly converged to the true number of states,  $S = 4$ . The posterior mean estimate was  $S = 4.0$  (SD=0.0) in both cases. Recovery of the parameters describing the location and shape of the ROIs was also accurate.

## RESULTS

In describing the results of the analysis of the data in Section 2, we focus on the results for the seven images that contain a nodule (Cases 1, 3, 9, 10, 11, 12, 14). Table 1 provides a description of the data. It shows for each of the seven cases the average gaze time on the image, on the image caption and on the nodule (in seconds), and the number of participants that identified the nodule correctly. Respondents spend from 28.3 (Case 11) to 69.3 seconds (Case 3) on the images, look at the image caption from about 2.0 (Case 9) to 5.4 seconds (Case 3), and spend between 5.0 (Case 11) and 18.1 (Case 3) seconds on the nodule. There appears not to be a clear relation between viewing times and the number of participants that find the nodule: if anything cases in which the nodule is found less often (Cases 3, 10, and 12) tend to have longer gaze times for the image and the nodule, which is in line with earlier findings (Krupinski, 1996; Nodine et al., 2002).

Figure 2 provides the scan-path pattern for all sixteen participants on Case 14, the image of which is shown in Figure 1, and which we will use throughout to illustrate the results. Participants searched between 40 seconds and almost 4 minutes for a nodule. Figure 2 illustrates the heterogeneity of the scan-path across participants. It also highlights

the need to use model-based inference to draw conclusions about search behavior for nodules from these data.

[Insert Table 1 and Figure 2 about here]

The estimates of the parameter  $\beta$  of the HMM, reflecting the influence of the luminance of regions in the image, are very similar across these seven images. Table 2 shows the posterior means and standard deviations of this parameter. The posterior mean of  $\beta$  is around 0.8 and the 95% credible intervals do not cover 0. The estimate varies relatively little across the images. This reveals that the brighter a region in the X-ray image is, the more visually salient it is, and the more likely it will attract eye movements. Note that in X-ray images, brighter regions are associated with more impenetrable structures (Figure 1). Because a nodule is usually brighter than its surroundings, participants will have a larger probability to fixate on it with their eyes. The estimates in Table 2 indicate that if a region is 10% brighter, the probability that participants' eyes fixate on it increases by about 8%. This underlines the potential of image enhancing methods that increase the brightness of a nodule relative to its surroundings, which will substantially enhance the probability that it is fixated on and identified.

[Insert Table 2 about here]

The average number of estimated ROIs across the seven X-ray images is 7.14 (SD=1.07). For each of the seven cases, the median numbers of ROIs were, Case 1: 8, Case 3: 8, Case 9: 6, Case 10: 8, Case 11: 8, Case 12: 6, and Case 14: 6. Figure 3 shows the estimated ROIs for all seven X-ray images with a nodule on it, as recovered from participants' eye movement data. Each panel in the figure plots the posterior means of  $\vec{\mu}_s$

and the 80% probability ellipse, calculated from the posterior mean of  $\Sigma_s$  for each state  $s$ . The real presence of the nodule on each image is indicated by a small solid circle. As predefined, the image caption is state 1, and the other ROIs are sorted in the order of their x-coordinates. Figure 3 reveals that each of the images not only has a different number of ROIs, but also that the sizes and shapes of these ROIs are substantially different. Some of the images have ROIs that are more circular in shape (e.g., Case 12), others have ROIs that are elongated (e.g., Case 1), while others have overlapping ROIs of different sizes and shapes (Case 10, 14). In addition, the direction of the ROIs can be more horizontal (Case 14) or vertical (Case 11), and the ROIs can be arranged top to bottom (Case 12), or left to right (Case 9). The number and shapes of the ROIs vary across images because the structures visible in chest X-rays vary considerably from patient to patient, which causes different eye movement patterns (Figure 2).

Importantly, we can see from the figure that six out of the seven images have a single well-defined ROI surrounding the nodule. This ROI is considerably smaller than the other ROIs without a nodule. Only Case 3 has a larger nodule that is covered by two ROIs. Thus, our model recovers the ROI covering a nodule quite well from eye movement data. Across all seven images, the size of the ROI that contains the nodule is significantly smaller than that of the other ROIs: 9, 559 pixels on average for nodule states and 54, 114 pixels for others: a difference of almost a factor six. Across all images, the number of participants that correctly finds the nodule is significantly larger for images for which the size of nodule state is smaller ( $p = 0.06$ ). This performance is remarkable, because the eye movements were collected from laypeople with virtually no experience in interpreting X-ray images, or searching for nodules on them. Indeed, the nodule state is recovered well even for images

where the majority of the participants did not find the nodule. For Cases 3 and 10, for example, only a single participant found the nodule, and in Case 12 only four found it. For none of the cases more than half of the participants found the nodule in the X-ray image (Figure 1), but still, the ROI containing the nodule was recovered well from the eye movement data in all those cases, using the HMM.

[Insert Figure 3 about here]

We take case 14 as an example (see Figures 1 and 2). For this case, 7 out of 16 participants found the nodule. We provide the posterior means and standard deviations of  $\vec{\mu}_s$  in Table 3 (notice that the y-axis is labeled from top to bottom). The table illustrates that each ROI is clearly separated from the others, and the small posterior standard deviations suggest that the eye movement data provide a considerable amount of information on the location of the ROIs. Table 3 also displays the posterior mean and standard deviation of the elements of  $\Sigma_s$ . This shows that there are substantial differences in the shape and orientation of the ROIs (see Figure 3), and that the nodule state 5 is the smallest in size. Not only is the size of the ROI that contains the nodule smallest, the posterior standard deviation of the estimates for this state (both location and shape parameters) is also substantially smaller than those of the other ROIs.

[Insert Table 3 about here]

Table 4 shows the posterior mean switching rate for each image across participants: the percentage of state changes throughout the search for the nodule, or in other words, the rate at which the eyes move between the identified ROIs. Switching rates are low, roughly between three and five percent, and the table shows that there are some

differences between images. Participants thus mostly move their eyes within the ROIs, and only occasionally switch to another ROI. This pattern corroborates previous findings of visual search in different contexts, which has also found that eye movements of participants are mostly confined to local regions, with an occasional longer jump of the eyes to redirect attention to other regions (Liechty, Pieters & Wedel, 2003; Van der Lans, Pieters, & Wedel, 2008).

[Insert Table 4 about here]

We represent the posterior means and standard deviations of the complete transition probability matrix  $Q_i$  in Figure 4, again using Case 14 as an example. In the figure, circles indicate the posterior mean estimates averaged across participants. The size of the circle is proportional to the square root of magnitude of that element of  $Q_i$ . Figure 4 contains separate plots for participants who found the nodule ("success"), who reported to have found the nodule, but did not ("false positive"), and for those participants who failed to find the nodule ("failure"). Notice that for Case 14 the nodule is in ROI 5. The left panel shows that the value of  $q(5, 5)$  is the largest. This suggests that once participants' eyes are in the ROI that contains the nodule they tend to stay fixated in that region, indicating that the nodule is correctly recognized. On the other hand, the center panel shows smaller values of  $q(5, 5)$  (success: 0.97, false positives: 0.75, failures: 0.93). This suggests that the participants that indicated the wrong location to contain the nodule have a higher probability to move to other ROIs even if they reach the ROI that contains the nodule. The region containing the nodule is inspected, but apparently the nodule is not recognized as such. Further, the right panel shows that participants that fail to find the nodule jump from

the image caption (ROI 1) to each of the other ROIs much more frequently. This may indicate that if they cannot find the nodule, they look for cues in the caption about whether or not the image in question should contain a nodule.

[Insert Figure 4 about here]

## DISCUSSION

Visual search is one of the tasks humans face most frequently, whether it is searching for packages on the shelves of supermarkets or on shopping web-sites, searching for traffic signs while driving, searching for suspicious objects in X-ray images at security checkpoints, or searching for nodules in medical images. Eye movement recordings provide detailed moment-to-moment insights into those visual search processes that are difficult to obtain otherwise (Zelinsky, 2008). Because eye movement recordings produce a large amount of data, statistical models are important tools for making inferences on the search process (Van der Lans, Pieters & Wedel, 2008). In this study, we have proposed a Hidden Markov Model to characterize the eye movements people make on unobserved Regions of Interest during visual search for nodules in chest X-ray images. Applications to search for nodules on other types of X-ray images, such as mammograms, are straightforward (Nodine et al. 2002). Yet, the model has the potential of much wider application to eye movement data in fields such as web-usability, engineering, marketing and psychology (Duchowski, 2002). The most compelling advance of the proposed model is the flexibility in identifying regions of interest from the eye movements themselves, both their number and their location and shape. In most eye movement research a key step in the analysis is the

definition of Regions of Interest: regions on images and other stimuli that the researcher hypothesizes to be relevant a priori and defines manually on the image (Holmqvist et al., 2013; Wedel & Pieters, 2008). The proposed HMM can be used to identify such areas of interest post hoc from the eye movement patterns themselves, which is particularly useful for unstructured stimuli, or research in which there are no strong prior expectations of what regions may attract attention. The model then helps to identify those regions on the image that are of interest to the participants, rather than being a priori thought relevant by the researcher. The application to chest X-rays indeed showed that the sizes, shapes and locations of the ROIs vary considerably within and between images. The proposed HMM model allows participants to be heterogeneous in their search behavior on these regions of interest. We used the luminance of each pixel in the image as a prior on the location of the regions of interest. This combines the ideas of image analysis with eye tracking analysis, and it turned out that luminance had a strong effect on the location of the Regions of Interest that contain a nodule. The model may need to be extended to accommodate the specific applications in other fields of research. For example, other image features beyond luminance, such as edges, contrast, and colors, can be included as prior information on the location of ROIs, extending the approach proposed here for one feature. Further, whereas in the present application the location and shape one of the Regions of Interest was a priori known, in other applications there may be a mix of a number of known and an unknown number of unknown regions of interest.

We applied the model to analyze eye movement data collected during the search for potentially cancerous nodules on chest X-rays. The search for pulmonary nodules is a very difficult task, also for experienced radiologists, who, even when the nodule is in retrospect

visible, may miss them up to thirty percent of the time (Krupinski, 2000). Recovering the nodule from only the X-ray image without human judgment proves to be very difficult, as current systems based on digital image analysis come up with as much as two or three false positives per image (Kakeda et al., 2004). Describing and analyzing human search for such nodules is therefore the focus of a growing area of inquiry that aims to improve the search effectiveness of radiologists by showing which search strategies work and which do not, and how errors arise (Krupinski, 2000). Applying the proposed model to eye movement data collected from a sample of participants that had no prior experience in searching chest X-ray images for nodules, we found an average of about seven regions of interest, that in some cases, partially overlapped. Participants mostly search within these critical regions, and only occasionally (up to five percent of the time) switched from one ROI to another. Using our model, we found one ROI that covers the nodule precisely for all images except for one (in which it was covered by two areas), and that ROI also had higher levels of associated posterior confidence. The smaller the size of that ROI was, the higher the accuracy of search on the image in question. But even if very few participants found the nodule, the HMM model identified a ROI that contained it. In addition, participants that are effective in finding the nodule revisit this area more frequently throughout their search. It is the combination of the use of features of the digital image and the eye movements of participants that allows us to identify the nodule with such high precision. This knowledge may help radiologists in developing more effective search strategies. Further, the results showed that areas on the images that are brighter than their surroundings pop out and attract more eye movements, which increased the chance that the nodule is detected and recognized. This reveals potential for the use of gaze contingent display to improve

radiologists' search effectiveness (Duchowski, Cournia, & Murphy, 2004). Gaze contingent displays optimize the amount and nature of visual information displayed in real time, based on the moment-to-moment location of the observer's recorded eye fixations. For example, through digital image processing, the luminance and contrast of candidate locations of nodules can be enhanced based on what the radiologist looks at each moment, and zoom and resolution can be enhanced in real-time. These tools hold promise to improve search by radiologists, and deserve further investigation. Thus, our analysis suggests feedback, training and real-time image enhancement strategies to improve the search for nodules in chest X-ray images by radiologists.

Our study has some limitations. First, we focused on the detection of only a single nodule, whereas in a real life scenario a radiologist is looking for all types of pathology, including abdominal disease, rib fractures, neck disease, disease of the spine and heart and blood vessels and so on. Second, in real life situations multiple nodules may be present. And third, our study focused on laymen as participants in the study, and did not investigate search by radiologists. Previous research has shown that laymen and trained radiologists do not fundamentally differ in their visual skills (Nodine & Krupinski, 1998), but differ in their level of training and experience. But, eye movement patterns by trained radiologists may be different from the ones revealed in this study. Future research may seek to replicate and extend the present findings to these contexts.

Nevertheless, the fact that we obtained these results from eye movement data collected on laymen with no prior experience in searching for these nodules is important. Even for images for which only very few respondents identified the nodule correctly, the

HMM accurately pinpointed the region containing the nodule based on their eye movements. If our results can be reliably replicated in other settings and for larger samples of chest X-ray images and respondents, this opens up the possibility of exploring crowdsourcing to aid in the detection of a limited number of regions of interest for closer inspection by radiologists. Crowdsourcing involves procedures to obtain solutions to problems by combining the efforts of a large number of people in online communities, rather than from a smaller number of experts. It has been successfully applied to search tasks in astronomy and medicine, amongst others (Doan, Ramakrishnan & Halevy, 2011). For the procedures discussed in this article to lend themselves to crowdsourcing, eye movement data need to be collected at a large scale in online settings. Recent developments in eye tracking technology make this possible. Rather than via more expensive infrared eye tracking cameras, eye movements can now be recorded remotely through webcams (Hiley, Redekopp, & Fazel-Rezai, 2006; Kim & Ramakrishna, 1999; Su, Su & Chen 2005). Spatiotemporal precision is as yet not as high as for infrared eye trackers, and there are challenges in ensuring data of sufficient quality (Hiley, Redekopp, & Fazel-Rezai, 2006). To some extent, the larger samples of participants in crowd-sourcing applications may offset this lack of precision. Crowdsourcing may be a promising route to improve the effectiveness of search for nodules in chest X-rays in lung cancer screening programs, through webcam based eye tracking and the HMM developed in this paper. We hope that the research presented in this paper provides a useful starting point for these developments.

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Figure 1. X-ray image with one nodule present (the headline is: "Case 14: 56 year old male with weight loss and coughing blood"). The nodule is visible as a bright roundish area in the upper-right quadrant on the 5<sup>th</sup> rib from the top.

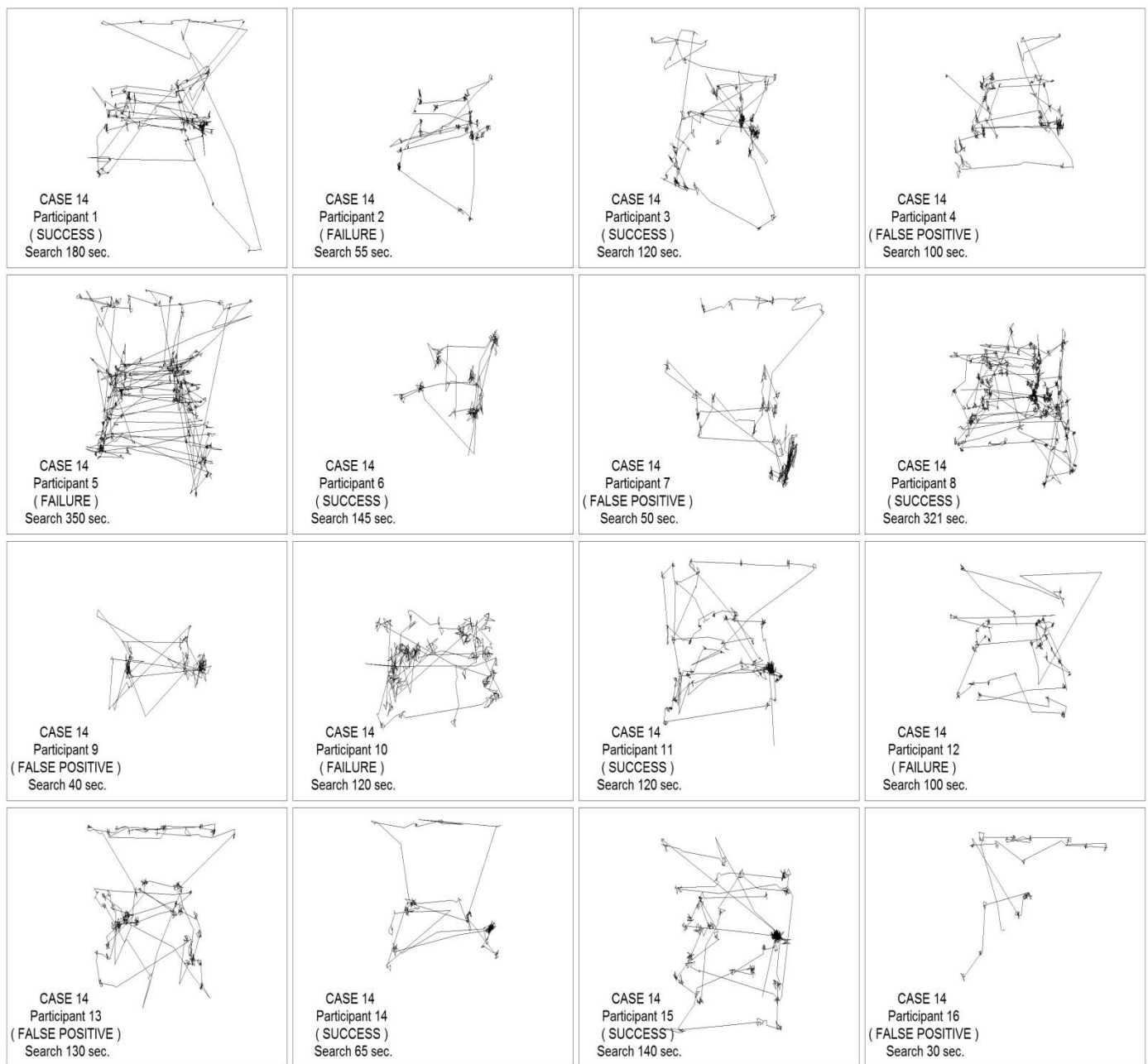


Figure 2. Scan-paths for 16 participants (Case 14). Out of sixteen participants, seven successfully find the nodule, four believe there is no nodule, and believe a nodule is in the wrong location.

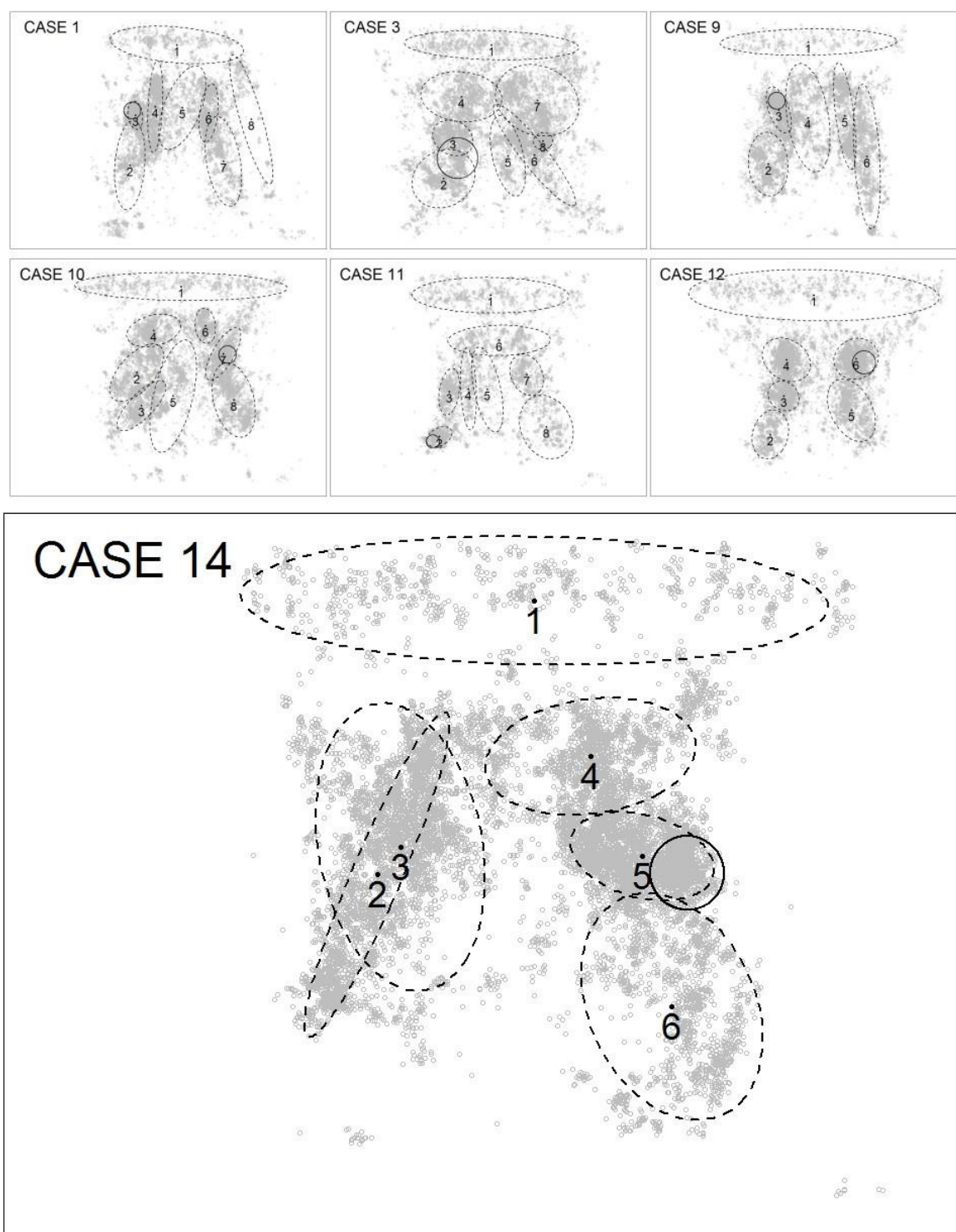


Figure 3. Estimated regions of interest on seven chest X-ray images with a nodule. The dashed ellipses are the estimated ROIs, with the state number indicated in center. The solid circles represent the location of the nodule.

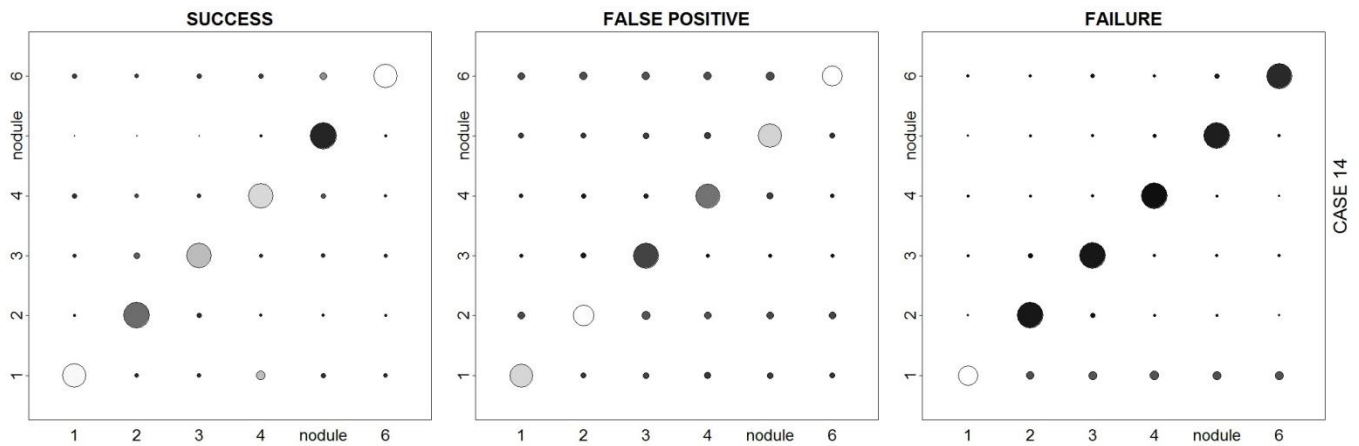


Figure 4. Plot of the transition probabilities for the three groups of participants for Case 14.

The size of the circle at  $(s, r)$  is proportional to the square root of the posterior mean of  $Q_i(s, r)$  averaged across participants in that group. The brightness of the circle is proportional to the average posterior standard deviation.

Table 1. Description of the data for each of the seven images that contains a nodule: average time on the image, the caption and the nodule (sec.), and number of participants that found the nodule.

|                        | Case<br>1 | Case<br>3 | Case<br>9 | Case<br>10 | Case<br>11 | Case<br>12 | Case<br>14 |
|------------------------|-----------|-----------|-----------|------------|------------|------------|------------|
| Time on image (sec.)   | 55.925    | 69.266    | 52.438    | 57.041     | 28.250     | 55.203     | 40.972     |
| Time on caption (sec.) | 3.719     | 5.466     | 1.991     | 4.188      | 4.125      | 5.325      | 2.813      |
| Time on nodule (sec.)  | 12.719    | 18.088    | 9.813     | 7.863      | 5.013      | 12.216     | 9.491      |
| Number finding nodule  | 7         | 1         | 7         | 1          | 8          | 4          | 7          |

Table 2. Posterior means and standard deviations (in parenthesis) of the luminance parameter  $\beta$  for the seven images that contain a nodule.

|      | Case 1 | Case 3 | Case 9 | Case 10 | Case 11 | Case 12 | Case 14 |
|------|--------|--------|--------|---------|---------|---------|---------|
| Mean | 0.765  | 0.809  | 0.724  | 0.827   | 0.840   | 0.739   | 0.718   |
| SD   | 0.221  | 0.341  | 0.396  | 0.218   | 0.172   | 0.311   | 0.319   |

Table 3. Posterior means and standard deviations (in parenthesis) of parameters for Case 14.

| Parameter    | Estimates | Parameter  | Estimates |
|--------------|-----------|------------|-----------|
| $\mu_1(x,y)$ | 529.3     | $\Sigma_1$ | 30783.2   |
|              | (5.7)     |            | (1507.3)  |
|              | 65.4      |            | 595.0     |
| $\mu_2(x,y)$ | (1.3)     | $\Sigma_2$ | 1477.5    |
|              | 361.3     |            | (221.9)   |
|              | (1.2)     |            | (115.1)   |
| $\mu_3(x,y)$ | 358.5     | $\Sigma_3$ | 1872.4    |
|              | (2.3)     |            | (65.1)    |
|              | 386.0     |            | -3893.8   |
| $\mu_4(x,y)$ | (1.6)     | $\Sigma_4$ | 9372.3    |
|              | 329.0     |            | (128.6)   |
|              | (2.4)     |            | (279.6)   |
| $\mu_5(x,y)$ | 589.5     | $\Sigma_5$ | 2547.3    |
|              | (1.6)     |            | (96.7)    |
|              | 232.2     |            | 912.8     |
| $\mu_6(x,y)$ | (0.9)     | $\Sigma_6$ | 7357.5    |
|              | 644.5     |            | (123.9)   |
|              | (0.7)     |            | (254.2)   |
| $\mu_7(x,y)$ | 339.0     | $\Sigma_7$ | 3933.7    |
|              | (0.5)     |            | (138.4)   |
|              | 676.4     |            | -382.3    |
| $\mu_8(x,y)$ | (1.6)     | $\Sigma_8$ | 1222.0    |
|              | 499.8     |            | (58.2)    |
|              | (2.2)     |            | (42.1)    |

Table 4. Posterior mean and standard deviations (in parenthesis) of the switching rates between ROIs for the seven images that contain a nodule.

|          | Case 1 | Case 3 | Case 9 | Case 10 | Case 11 | Case 12 | Case 14 |
|----------|--------|--------|--------|---------|---------|---------|---------|
| Mean (%) | 4.6    | 4.9    | 2.9    | 4.1     | 5.0     | 3.5     | 3.4     |
| SD       | (2.2)  | (2.5)  | (1.3)  | (1.2)   | (3.0)   | (1.1)   | (1.4)   |